

Simulation and Modeling

Lecture 10

Modeling Complex Systems

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Multi-teller Bank with Jokeying

Problem Statements

- A bank with 5 tellers
- Opens its door at 9:00 am and closes at 5:00 pm
 - Operates until all customers arrived by 5:00 pm are served
- Inter-arrival times of the customer are IID exponential random variables with mean 1 minutes
- Service times of customers are IID exponential random variables with mean 4.5 minutes
- Each teller has a separate queue
- Arriving customer joins the shortest queue
 - Leftmost shortest queue in case of a tie

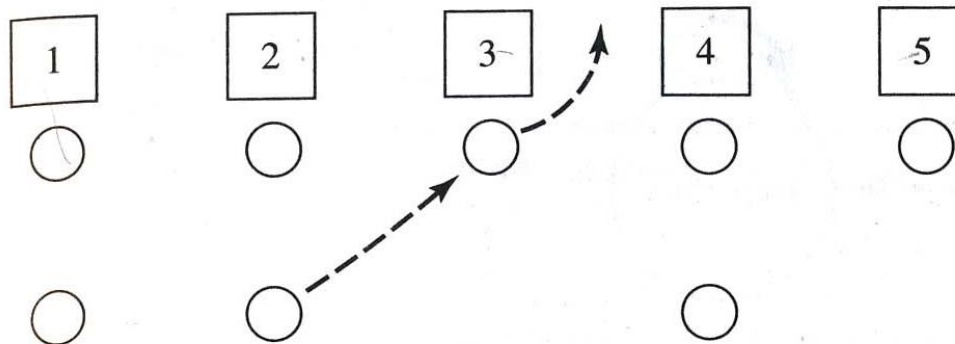
Problem Statements

- Let n_i denote the total customers in the teller i
- If completion of a service at teller i causes

$$n_j > n_i + 1$$

for teller j , then the last customer of j jockeys (i.e., moves) to the tail of teller i

- If jockeying is feasible for more than one customer
 - Closest leftmost customer jockeys



Simulation Objectives and Goals

- Management is concerned with
 - Operating cost
 - Quality of service
- Changing the number of teller
 - Does the bank need an additional teller?
 - Is it possible to decrease the number of teller?

Conceptual Model

- State Variables:
 - $x_i, i = 1, 2, \dots, n$: teller i 's status (busy/idle)
 - $q_i, i = 1, 2, \dots, n$: number of waiting customers at teller i
- Set of events
 - $E = \{a, d, t\}$
 - Arrival of a customer (arrival event)
 - Departure of a customer at teller i (departure event)
 - Closing of the bank (termination event)

Conceptual Model

- Arrival Event
 - If all servers are busy
 - Increases the # of waiting customers in the leftmost shortest queue,
 - If at least one server is idle
 - Changes the status of the leftmost idle server from idle to busy

Conceptual Model

- Departure Event
 - If the queue at the departing customer's server is non-empty
 - Decreases the number of waiting customer in the queue
 - If the queue at the departing customer's server is empty
 - Changes the status of the teller (server) from busy to idle

Conceptual Model

- Termination Event
 - Restrict further arrival of customer
 - Remove the arrival event from the event queue
 - Ensures further arrival

Conceptual Model

- Input Variables
 - Customer inter-arrival times
 - Customer service times
 - Bank closing time
- Output Variables
 - Expected time-average number of customers in each queue
 - Expected average delay in each queue
 - Expected maximum delay in each queue
 - Expected teller utilization

Specification Model

- State Equation

$$x_i(t^+) = \begin{cases} x_i(t) + 1, & \text{if arrival occurs and leftmost idle teller is } i \\ x_i(t) - 1, & \text{if departure occurs and } q_i = 0 \text{ for all tellers} \\ x_i(t), & \text{otherwise} \end{cases}$$

$$q_i(t^+) = \begin{cases} q_i(t) + 1, & \text{if arrival occurs and leftmost shorted queue is } i \\ q_i(t) - 1, & \text{departure at } i \text{ th teller } q_i > 0, q_i \geq q_j \forall j, i \neq j \\ q_i(t), & \text{otherwise} \end{cases}$$

Specification Model

- State Space
 - $X_i = \{0,1\}$, for $i = 1, 2, \dots, n$
 - $q_i = \{0,1, 2, \dots\}$, for $i = 1, 2, \dots, n$
 - $S = \{(0,0), (0,0), (0,0), \dots$
 $(1,0), (0,0), (0,0), (0,0), \dots$
 $(1,0), (1,0), (0,0), (0,0), \dots$
 $\dots \dots$
 $\}$

Specification Model

- Feasible Event Sets
 - $E = \{a, d, t\}$
 - $F_x \rightarrow a$ – always until the closing time
 - d – if X_i is busy
 - t – on or before closing time

Specification Model

- Output Equations

- Time average total number of customer
- $Q_i(t) \rightarrow \#$ of customer in queue i at time t
- Total waiting customer at t , $Q(t)$

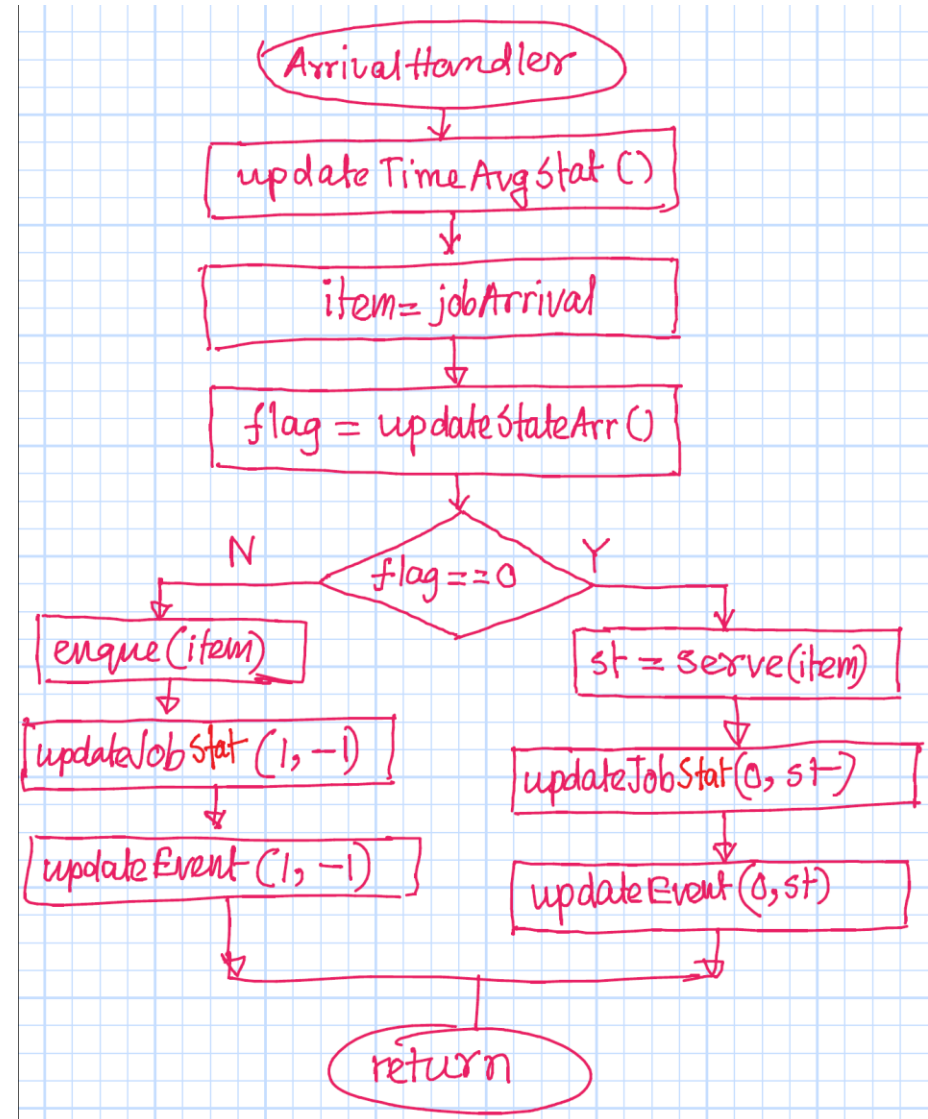
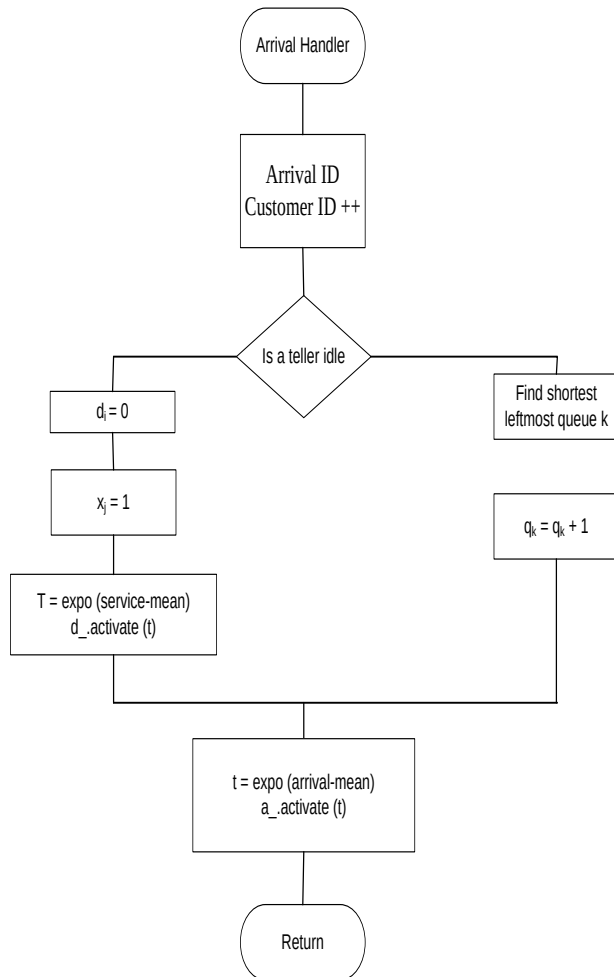
$$Q(t) = \sum_{i=1}^n Q_i(t)$$

$$\begin{aligned} - \hat{q} &= \frac{\int_0^T Q(t)dt}{T} = \frac{\int_0^T \sum_i^n Q_i(t)dt}{T} \\ &= \frac{\int_0^T Q_1(t)dt}{T} + \frac{\int_0^T Q_2(t)dt}{T} + \dots + \frac{\int_0^T Q_n(t)dt}{T} \\ &= \hat{q}_1 + \hat{q}_2 + \dots + \hat{q}_n \end{aligned}$$

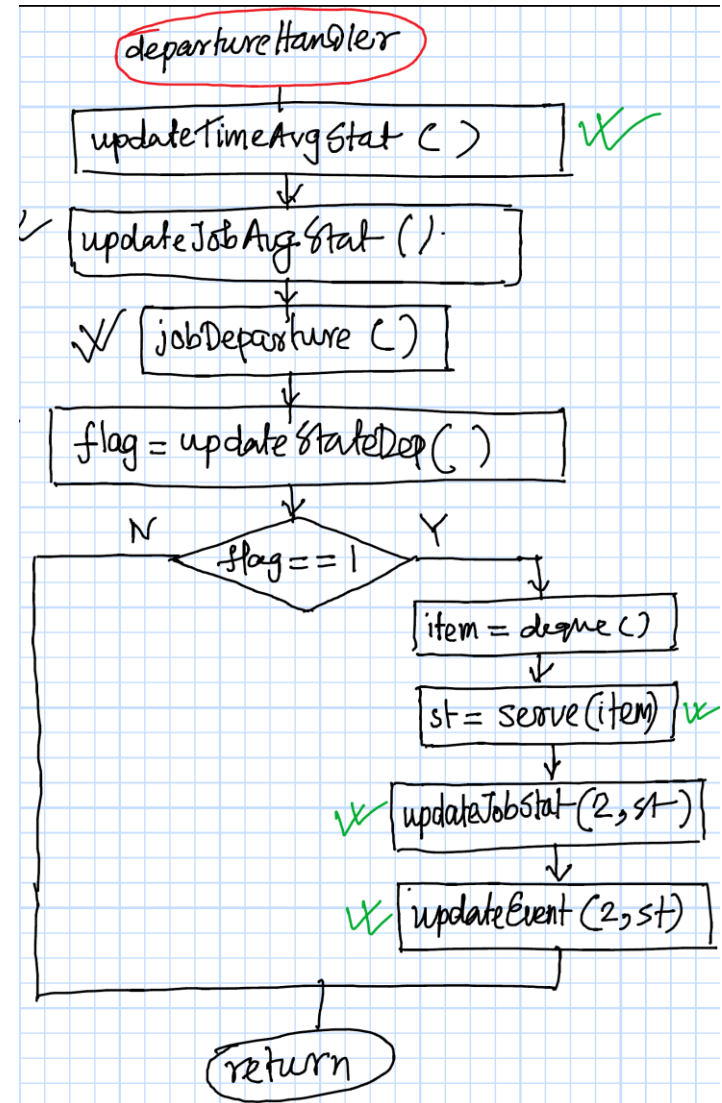
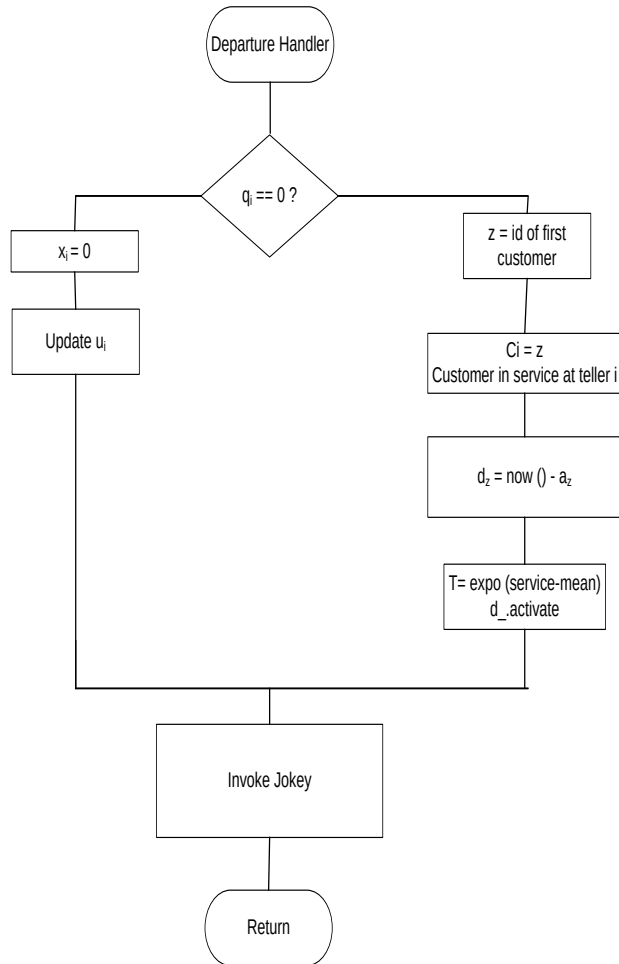
Specification Model

- Average delay in queue
- Let d_i denote queueing delay of the i th customer
- $d_i = b_i - a_i$
 - a_i is the arrival time of the i th customer
 - b_i time at which service of i th customer starts
- Let $\hat{d}$ denote the average queueing delay for the n customers
- $\hat{d} = \frac{\sum_{i=1}^n d_i}{n}$

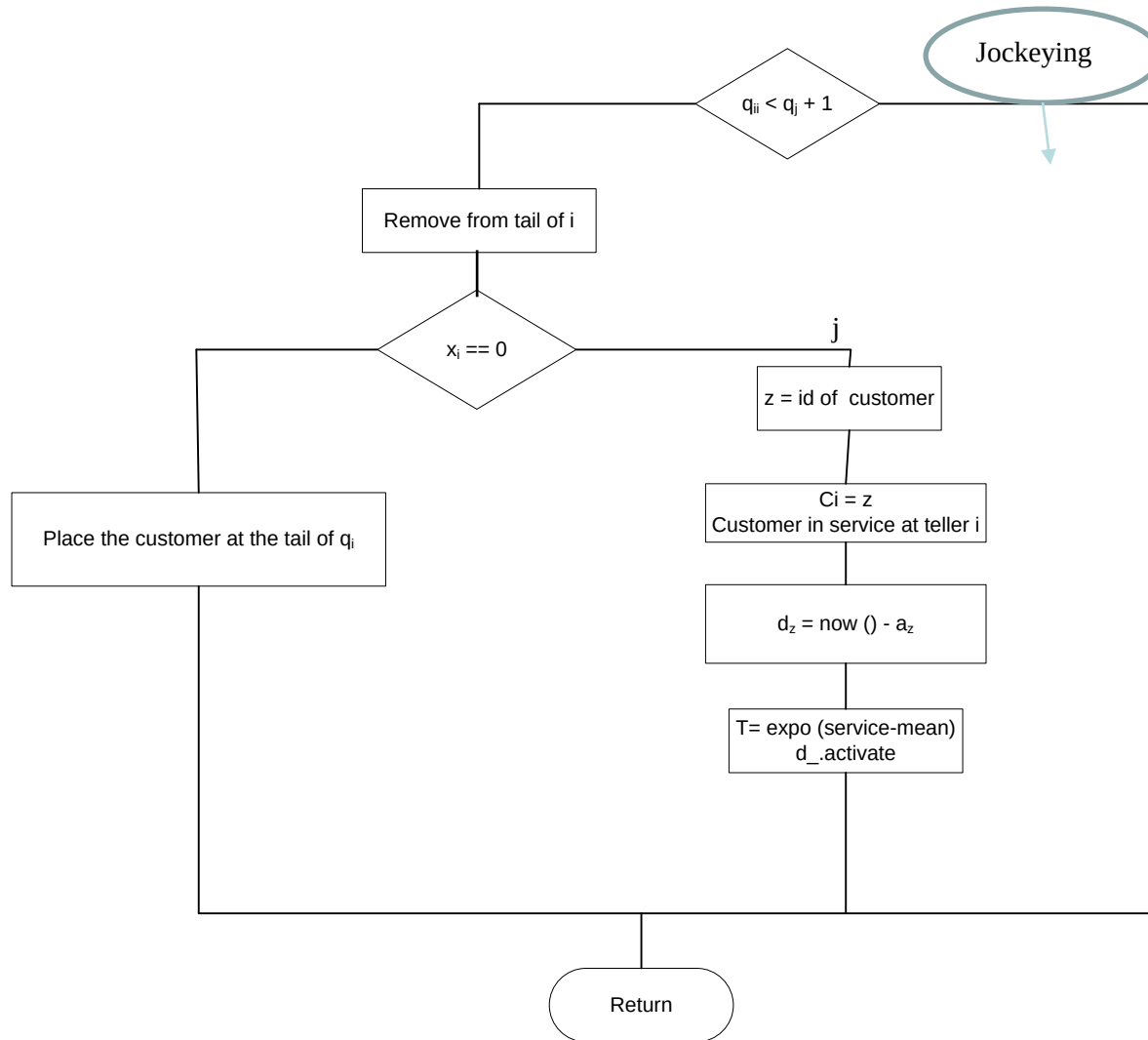
Computation Model: Arrival Handler



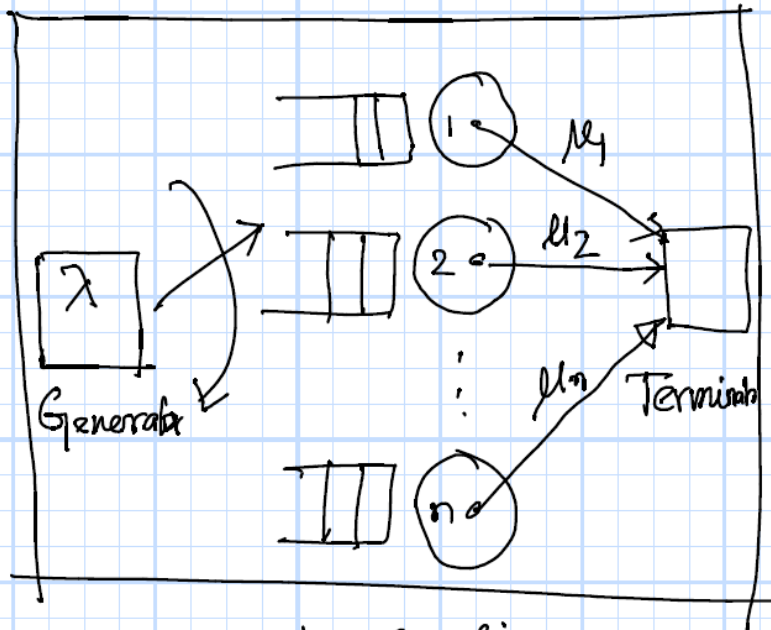
Computational Model: Departure Handler



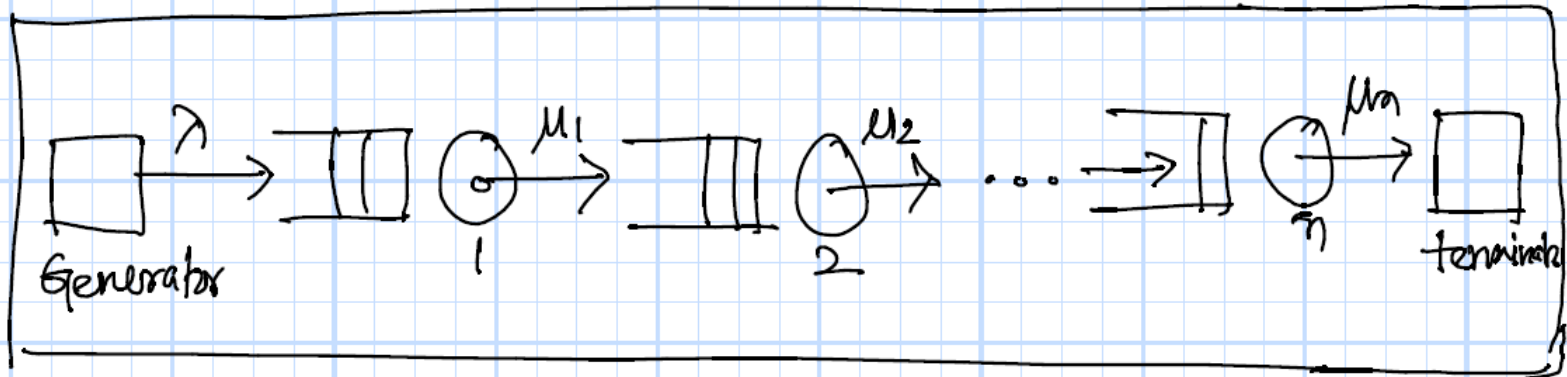
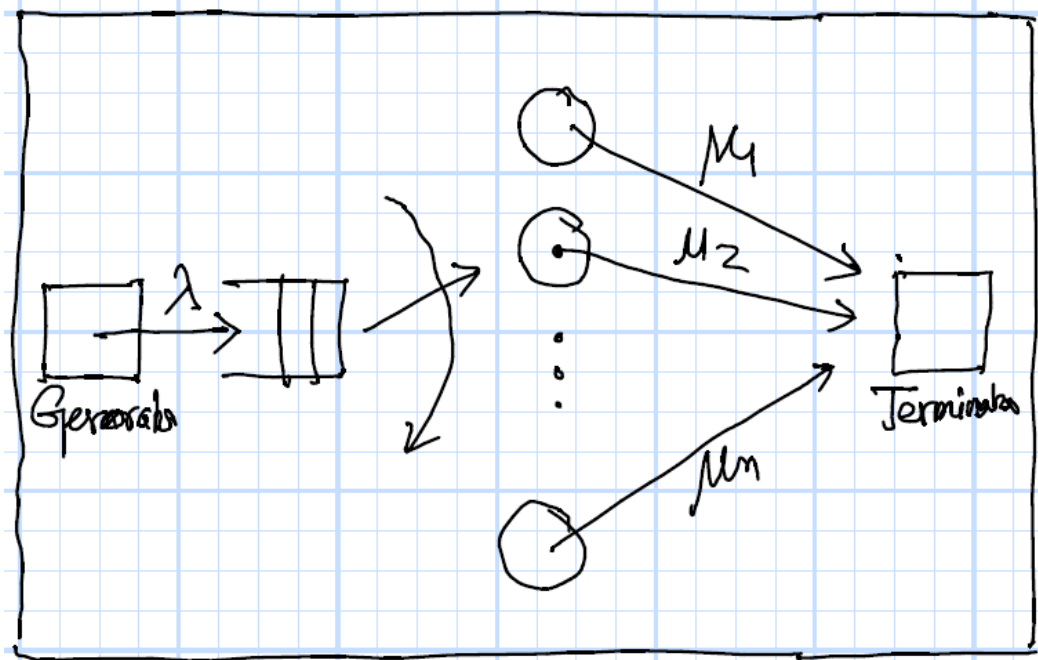
Computation Model



Multi-Server Queueing Systems



Service Station



Service Station

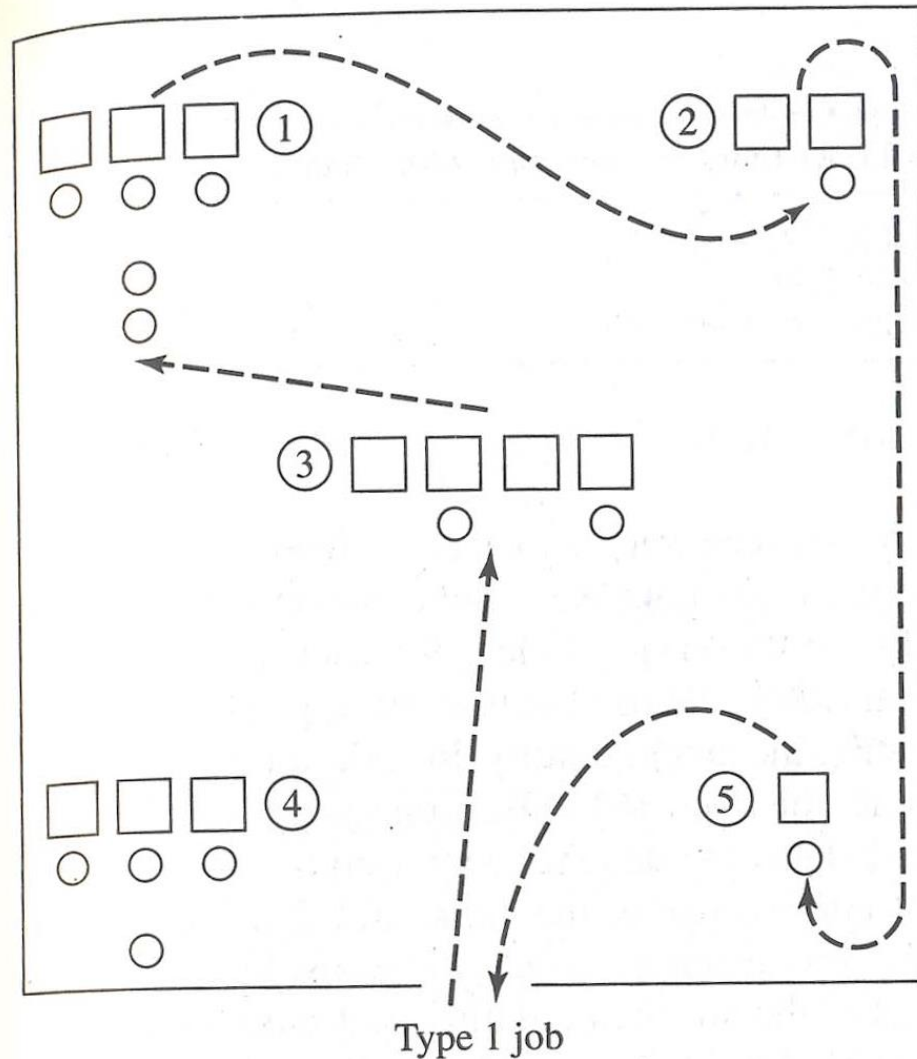
Job-Shop Model

(Model of a Manufacturing Systems)

Problem Statement

- Manufacturing systems consists of 5 workstations
- Stations 1, 2, ..., 5 consists of 3, 2, 4, 3, 1 identical machines, respectively
- Each station has a single queue
- The system, therefore, is a network of 5 multi-server queues
- Jobs arrival with interarrival times $\sim$ IID expo (0.25h)
- Jobs are of three types
 - Type 1 with Probability 0.3
 - Type 2 with Probability 0.5
 - Type 3 with Probability 0.2

Problem Statement



Problem Statement

- Job 1 requires 4 workstations
 - 3, 1, 2, 5
- Job 2 requires 3 workstations
 - 4, 1, 3
- Job 3 requires 5 workstations
 - 2, 5, 1, 4, 3
- Newly arriving job travel the system from one station to the next in the above order
 - At any work station, if all the machines are busy, the arriving job wait in the queue

Problem Statement

- Service time at a station is 2-Erlang random variables, whose mean depends on
 - Job type
 - The station
 - $[X = Y_1 + Y_2, r_1 = r_2 = r/2]$
- Mean service times for the jobs are
 - Job1 – 0.5(3), 0.6(1), 0.85(2), 0.5(5)
 - Job2 – 1.1(4), 0.8(1), 0.75(3)
 - Job3 – 1.2(2), 0.25(5), 0.7(1), 0.9(4), 1.0(3)

Simulation Objective and Goals

- To run the system for 365 8-hour consecutive days to measure/estimate
 - Average total delay in queue for each job (excluding service times)
 - Expected overall average job total delay
 - Use job type probability 0.3, 0.5, 0.2
 - Expected average number in queue
 - Expected average utilization of the machines in each station
 - Average delay in queue for each station
- Assuming all machines cost approx. same, and the opportunity to purchase one new machine to improve efficiency
 - What additional simulation run do we need
 - These new simulation will have 14 machines
 - Overall average job-total delay

Conceptual Model

- State Variables
 - $n_i \rightarrow$ # of busy machines at station $i, i = 1, 2, \dots, 5$
 - $n_i = \{0, 1, \dots, N_i\}$
 - N_i represents the number of machines at station i
 - $q_i \rightarrow$ length of queue at station $i, i = 1, 2, \dots, 5$
- Set of Events
 - Arrival of a job to the system
 - Arrival event (a)
 - Departure of a job from a workstation
 - Department event ($d_i, i = 1, 2, \dots, 5$)
 - End of simulation
 - Termination event (t)

Conceptual Model

- Arrival Event
 - Let the first STA for the job is the i th station
 - Arrival increases q_i by 1, if $n_i = N_i$
 - Else, it increases n_i by 1
- Departure Event
 - Let the departure occurs at station i
 - If $q_i > 0$, the departure decreases q_i by 1
 - Else, departure decreases n_i by 1
 - Also, a departure changes the status of the next station, if current station is not the last station for the job
- Termination Event
 - Restricts further arrival
 - Remove the arrival event from the event queue

Conceptual Variables

- Input Variables
 - Interarrival times
 - Job Types
 - Service times at different stations for different job types
- Output Variables
 - Delay in queue at station i th station
 - $d_i^q, i = 1, 2, \dots, 5$
 - Delay in queue for j th job
 - $d_j^q, j = 1, 2, 3$
- Server Utilization at each workstation
 - $u_i, i = 1, 2, \dots, 5$

Specification Model

- State Equation

- $$n_i(t^+) = \begin{cases} n_i(t) + 1, & \text{arrival occurs \& first station is } i \text{ and } n_i < N_i \\ n_i(t) - 1, & \text{departure occurs \& next station is } i \text{ \& } n_i < N_i \\ n_i(t), & \text{departure occurs at station } i \text{ \& } q_i = 0 \\ n_i(t), & \text{otherwise} \end{cases}$$

- $$q_i(t^+) = \begin{cases} q_i(t) + 1, & \text{arrival occurs \& first station is } i \text{ and } n_i = N_i \\ q_i(t) - 1, & \text{departure occurs at } j(j \neq i) \text{ \& next station is } i \\ q_i(t) - 1, & \text{departure occurs at station } i \text{ \& } q_i = 0 \\ q_i(t), & \text{otherwise} \end{cases}$$

- State Space

- $$n_i = \begin{cases} 0, \dots, 3, & i = 1 \\ 0, \dots, 2, & i = 2 \\ 0, \dots, 4, & i = 3 \\ 0, \dots, 3, & i = 4 \\ 0, 1, & i = 5 \end{cases} \quad q_i = 0, 1, 2, \dots$$

Specification Model

- Set of Events
 - $E = \{a, d_1, d_2, d_3, d_4, d_5, t\}$
- Feasible events
 - $F_x = a$ is always feasible
 - d_i if $n_i > 0$
 - t on or before closing time